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Incorporation of hydrogen bonding interactions with physical crosslinking to endow applicable mechanical property for microcrystalline cellulose-*graft* bottlebrush copolymer elastomers

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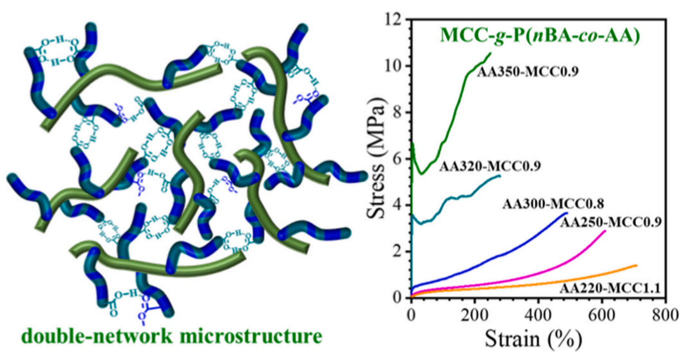
HIGHLIGHTS

- Microcrystalline cellulose-*g*-poly(*n*-butyl acrylate-*co*-acrylic acid) copolymers are synthesized by ARGET ATRP and hydrolysis strategy.
- MCC-*g*-P(*n*BA-*co*-AA) copolymers show the highest ultimate tensile strength of 10.6 MPa at an elongation at break of 249%.
- The synergistic effect of a double-network microstructure endows the MCC-*g*-P(*n*BA-*co*-AA) copolymers with strong mechanical properties.
- ARGET ATRP overcomes the limitations of acrylic acid copolymerization, ensuring well-defined grafts as compared to the RAFT polymerization.

GRAPHICAL ABSTRACT



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ABSTRACT

A series of microcrystalline cellulose-*graft* bottlebrush copolymer elastomers (microcrystalline cellulose-*graft*-poly(*n*-butyl acrylate-*co*-acrylic acid) (MCC-*g*-P(*n*BA-*co*-AA)) have been successfully synthesized via combining activators regenerated by electron transfer for atom transfer radical polymerization (ARGET ATRP) and hydrolysis reaction. MCC-*g*-P(*n*BA-*co*-*t*BA) copolymers, serving as the precursors of MCC-*g*-P(*n*BA-*co*-AA) copolymers can be dissolved in tetrahydrofuran (THF) to form solutions, whilst MCC-*g*-P(*n*BA-*co*-AA) copolymers can only form sols when dissolved in THF. MCC-*g*-P(*n*BA-*co*-*t*BA) copolymers are viscous and cannot be shaped, whilst MCC-*g*-P(*n*BA-*co*-AA) copolymers can be shaped due to higher thermal stability and tunable higher glass transition temperatures through adjusting the feed ratio of *n*BA and *t*BA monomers. It is found that microcrystalline cellulose functions as physical crosslinking points to form the primary network and the grafted P(*n*BA-*co*-AA) side chains form hydrogen bonding to build up the secondary network, thus, the dense double-network microstructure synergistically enhances the mechanical property of MCC-*g*-P(*n*BA-*co*-AA) copolymers for potential applications. With increasing PAA mass content, the mechanical property of MCC-*g*-P(*n*BA-*co*-AA) copolymers can be conveniently tuned from ductile one to robust one with a wide range of ultimate tensile strength from 1.4 to 10.6 MPa. Atomic force microscopy and small-angle X-ray scattering measurements reveal that the

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